



Q3	Map the breach to Cyber Kill Chain stages / MITRE ATT&CK	5 Marks
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Objective: Place each attacker action into the correct Cyber Kill Chain stage and, where possible, tag it with a MITRE ATT&CK technique ID.

Step-by-Step Algorithm

Step	Phase	Action	Command / Tool	What You Should See
1	SETUP	List the seven Cyber Kill Chain stages as your framework.	Cyber Kill Chain	Stage list ready
2	ANALYSE	Assign each attacker action to the stage where it belongs.	Kill chain mapping	Action per stage
3	ANALYSE	Look up a matching ATT&CK technique for each action.	ATT&CK Navigator	Technique IDs (T####)
4	TEST	Mark the earliest stage where detection was realistically possible.	—	Detection point flagged
5	REPORT	Record the mapping as a table for the report.	Table template	Kill chain table

Phase key: ● SETUP · ● CONFIGURE · ● EXECUTE · ● TEST · ● ANALYSE · ● REPORT · ● FINISH

Worked Example (with sample values & output)

Kill chain and ATT&CK mapping (Acme HealthCare)

Kill Chain stage	Observed action	ATT&CK technique
Reconnaissance	Employee emails researched	T1591
Delivery	Phishing email sent	T1566
Exploitation	Malicious attachment opened	T1204
Installation	Ransomware dropped	T1486
Actions on Objectives	Data exfiltrated	T1041



Earliest realistic detection: the Delivery stage. An email gateway scanning attachments could have quarantined the phishing message before it ever reached the user.

Reusable Template

Operation Iron Lattice—maps a targeted corporate espionage campaign against a critical infrastructure supplier into the 7 Cyber Kill Chain stages developed by [Lockheed Martin](#), complete with corresponding MITRE ATT&CK technique IDs.

Kill Chain stage	Observed action	ATT&CK technique (optional)
Reconnaissance	OSINT gathering & Infrastructure Scans: The threat actors map out the target supplier's public footprint, scrape employee profiles on professional networking sites for spear-phishing targets, and scan internet-facing assets for unpatched vulnerabilities.	<i>MITRE ATT&CK:</i> Active Scanning (T1595)
Delivery	Spear-phishing & Supply Chain Infiltration: The weaponized email is delivered directly to mid-level procurement engineers. Alternatively, a compromised trusted vendor's website is turned into a watering hole to catch target engineers downloading system schematics.	<i>MITRE ATT&CK:</i> Phishing: Phishing Attachment (T1566.001)
Exploitation	User Execution & Flaw Triggering: The victim opens the document, triggering an embedded vulnerability (e.g., a Microsoft Office remote code execution flaw) or enabling a macro that executes a silent PowerShell script to bypass endpoint controls.	<i>MITRE ATT&CK:</i> Exploitation for Client Execution (T1203)
Installation	Persistence & Tool Deployment: The payload deploys a lightweight backdoor into the local system and alters registry	•  Boot or Logon Autostart Execution:

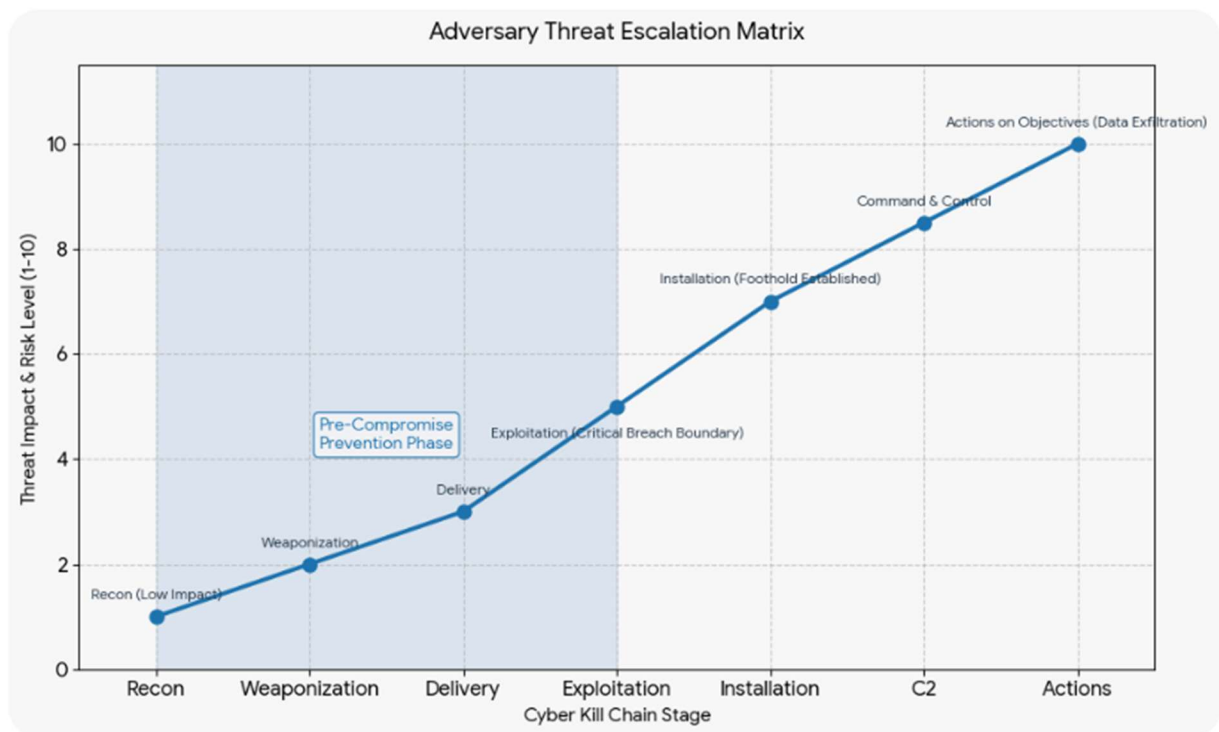


	run keys to ensure it remains active even if the machine reboots, establishing a permanent foothold on the IT network.	Registry Run Keys / Startup Folder (T1547.001) or Scheduled Task/Job (T1053) .
Actions on Objectives	. Persistence & Tool Deployment: The payload deploys a lightweight backdoor into the local system and alters registry run keys to ensure it remains active even if the machine reboots, establishing a permanent foothold on the IT network.	Exfiltration Over C2 Channel (T1041) or Data Encrypted for Impact (T1486)

Operation Iron Lattice represents a targeted corporate espionage campaign designed to compromise a critical infrastructure supplier, exfiltrate intellectual property, and establish long-term persistence within both corporate (IT) and operational technology (OT) environments.

Threat Progression Visualisation

The visual model below tracks the escalation of adversary capability against the timeline of Operation Iron Lattice. The blue area highlights the critical "**Break-In**" boundary. Stopping the attacker before or at Stage 4 (Exploitation) completely nullifies the financial and structural impact of the breach.



1. **Lockheed Martin Cyber Kill Chain (Strategic Level):** This framework tracks the phase-by-phase chronological progression of the attack. It is highly effective for executives and incident response managers to identify *where* an attacker is currently positioned in the attack life cycle.



2. **MITRE ATT&CK (Tactical Level):** While the Kill Chain gives the timeline, MITRE ATT&CK provides the specific technical granular actions. It answers *how* the attacker is achieving each phase (e.g., using PowerShell or modifying Registry Run Keys).
3. **The Diamond Model (Operational Level):** This map charts every individual malicious event as a four-point diamond: **Adversary, Capability, Infrastructure, and Victim**. Connecting these diamonds tracks how the adversary pivots across assets over time.

